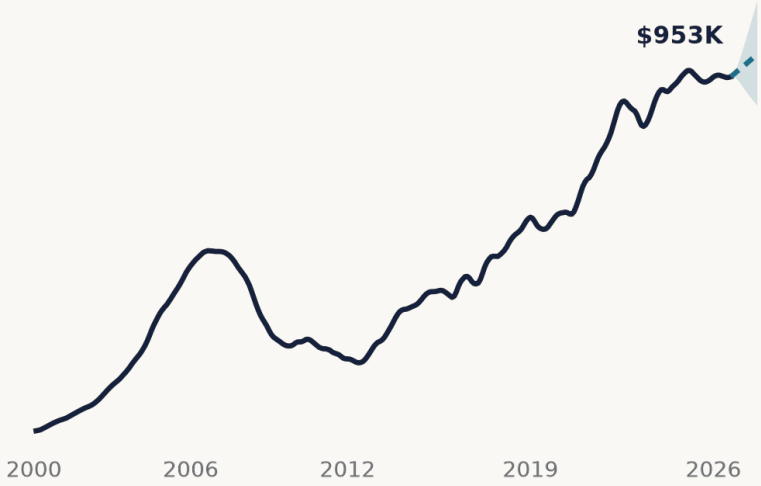
MGMTMSA 437 · FINAL PROJECT · OCTOBER 2026

Forecasting LA Home Values

Mortgage rates more than doubled after 2021. Did LA home values get harder to predict?

Yang Liu · Jinwoo Roh
Yiu-Hon Lee · Thanapol Doungsaeng

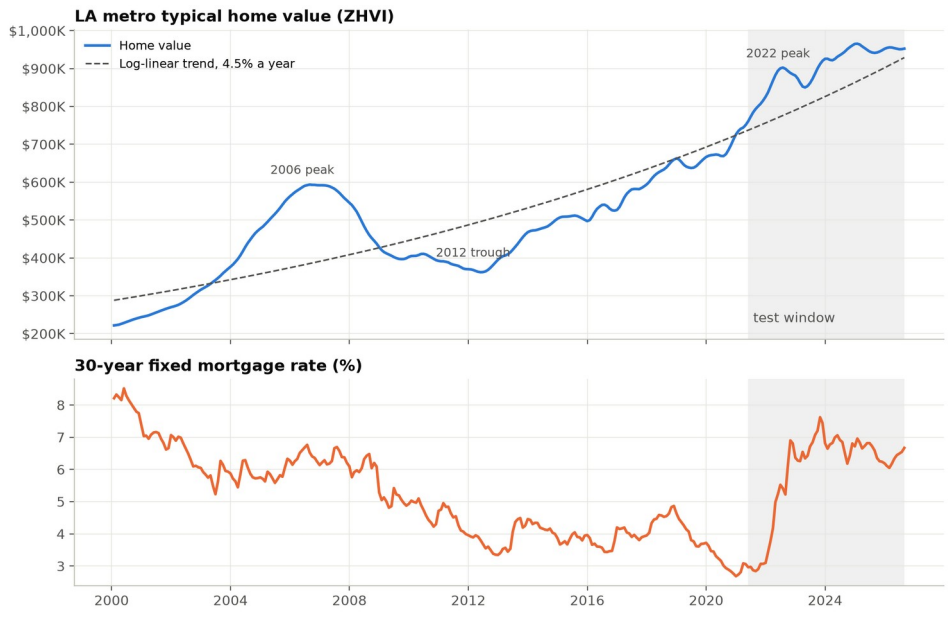
Zillow Home Value Index, LA metro, Jan 2000 to Aug 2026, plus 499 neighborhoods and the 30-year mortgage rate



Solid: actual · Dashed: our forecast to Aug 2027

THE QUESTION

LA values grew 5.6% a year for 26 years, then stalled when rates hit 7%



$952,601

typical LA-metro home, August 2026

-1.3%

vs the January 2025 peak of $965,634

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

DATA AND DESIGN

We tested 14 models two ways, so no model could win by luck

DATA

Zillow ZHVI, LA metro
320 months, Jan 2000 to Aug 2026

499 LA neighborhoods
Zillow neighborhood index

Mortgage rate and payroll jobs
FRED and BLS, LA and Orange County

14 MODELS + 3 LLMS

Baseline and classical
Naive drift · ARIMA · Holt · Prophet ×2

ML, deep learning, AI
XGBoost · LightGBM · LSTM · GRU · Chronos

Outside information
ARIMAX ×2 · VAR · factor VAR

Plus 3 chat LLMs
Claude · GPT · Gemini

2 TESTS

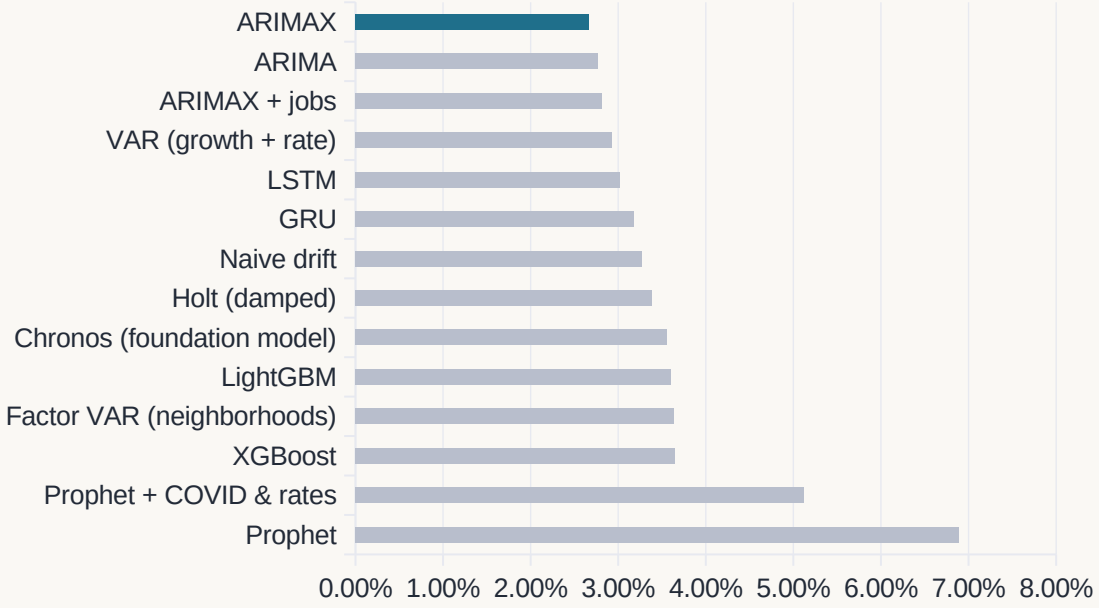
Blind 80/20
Train to Apr 2021, forecast 64 months in one shot

Rolling origin
Refit every 3 months, forecast 12 months ahead, 18 times

WHY NOT 1-MONTH ERRORS Zillow smooths this index, so every decent model is about 0.1% off one month ahead. We judge models at 6 to 12 months instead.

RESULT 1 · ROLLING TEST

In the fair test, ARIMA-family models beat machine learning



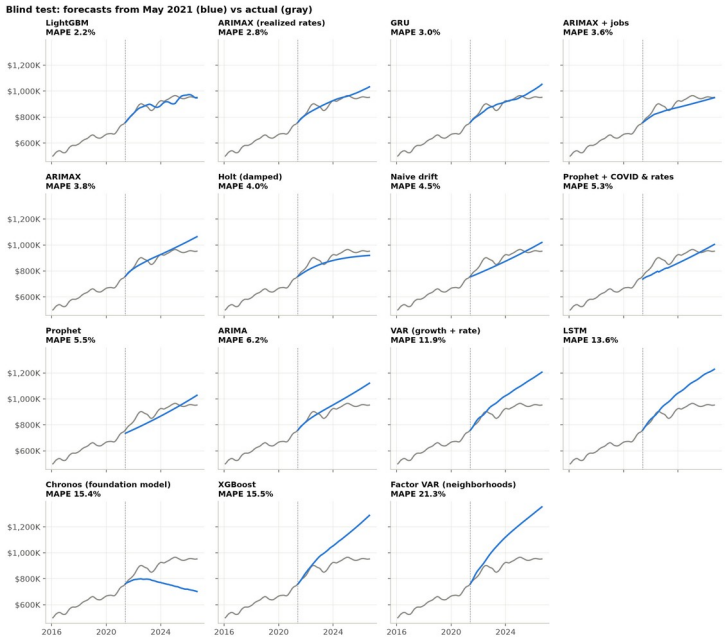
LOWEST ERROR

2.67%

ARIMAX, average error 1 to 12 months ahead. ARIMA 2.77%. LSTM 3.02%, XGBoost 3.65%, Prophet 6.89%. Also 1st on RMSE ($35.8K).

RESULT 2 · TEST DESIGN

One test window crowned LightGBM; 18 windows ranked it tenth



SAME MODEL, TWO TESTS

1st → 10th

LightGBM won the single 64-month test (2.2% error) but ranked 10th across 18 rolling tests: it got lucky in the one window that counted.

RESULT 3 · AI FORECASTS

Chat LLMs beat every model, until we hid which years they were forecasting



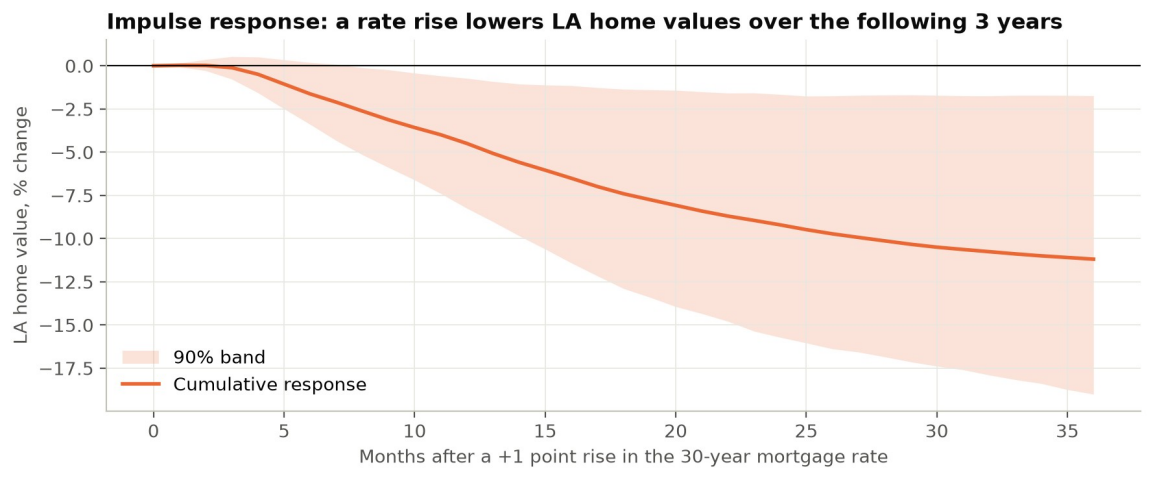
SAME MODEL, DATES HIDDEN

1.3 → 2.8%

Claude's error once it could not tell the place, dates or mortgage rate (ARIMAX 2.67%). All test dates are inside the LLMs' training data.

RESULT 4 · MORTGAGE RATES

A 1-point rate rise means LA values about 9% lower two years later



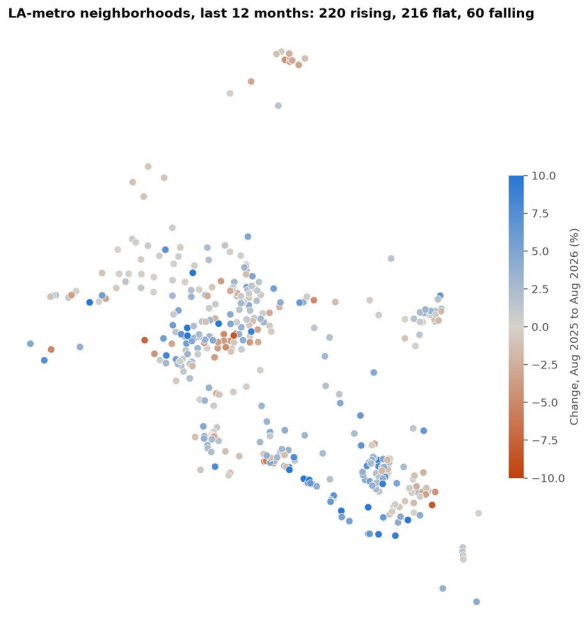
RATES WORK SLOWLY

−9.2%

after 24 months (−4.5% after 12). Rates help predict prices, not the reverse (Granger p = 0.05 vs 0.51). Local jobs add nothing once 2020–21 is set aside.

NEIGHBORHOODS · LIVE MAP

The metro average hides a split market, neighborhood by neighborhood



220 / 216 / 60

neighborhoods rising / flat / falling more than 2%, last 12 months

152 of 496

still below their 2022 peak, mostly in LA County

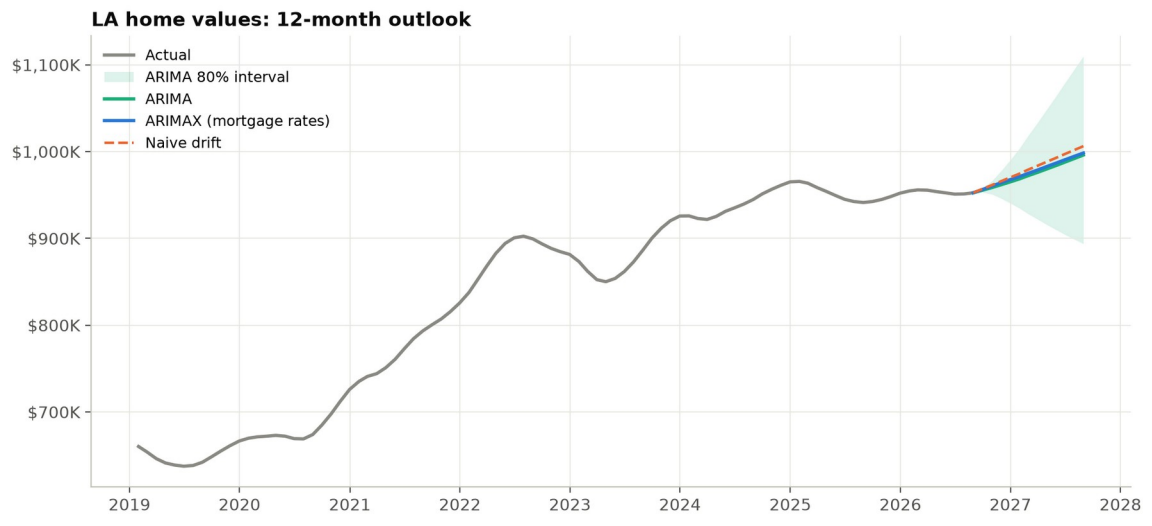
−1.2 pts

deeper 2022 drop per doubling of price in LA County (reverses in Orange County)

Live map: neiltd.github.io/la-housing-forecast

OUTLOOK

Our forecast: about 5% higher by August 2027, with a wide range



$998,391

ARIMAX forecast, Aug 2027 (+4.8%)

-6% to +17%

80% range for the next 12 months

+4.8% / +6.5%

median neighborhood forecast, LA County / Orange County

TAKEAWAYS

What we learned

01 **Simple beats complex for a smooth monthly index**

ARIMA-family models beat LSTM, GRU, boosted trees and the Chronos foundation model.

02 **How you test decides who wins**

One blind window ranked LightGBM 1st; 18 rolling windows, 10th.

03 **An LLM can't be backtested on years it has read about**

Claude, GPT and Gemini beat every model until we hid the dates; then they matched ARIMAX at best.

04 **Mortgage rates matter, slowly; local jobs don't add**

A 1-point rise lowers values about 9% over two years. Turning points stay hard to call.

05 **The metro average hides a split market**

Separate models per neighborhood beat one shared factor (2.8% vs 4.1% error).